

# Linear Algebra and Calculus Lecture 9

Eric Sandin Vidal

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## Inverse of a Matrix

With usual numbers we have multiplicative inverses such that  $aa^{-1} = 1$ , and with matrices the role of 1 will be played by  $I_n$ , which is the standard matrix of the transformation  $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  given by  $T(x) = x$ .

**Definition.** An  $n \times n$  matrix  $A$  is said to be **invertible** if there exists an  $n \times n$  matrix  $C$  such that

$$\begin{aligned}AC &= I_n \\CA &= I_n.\end{aligned}$$

The matrix  $C$  is called the **inverse** of  $A$ . If an  $n \times n$  matrix is not invertible, then it is called **singular**, otherwise it's **nonsingular**.

**Example.** Consider  $A = \begin{pmatrix} 2 & 5 \\ -3 & -7 \end{pmatrix}$  and  $C = \begin{pmatrix} -7 & -5 \\ 3 & 2 \end{pmatrix}$ , then  $AC = I_2$  and  $CA = I_2$ .

More generally, we have

**Theorem.** Consider  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . If  $ad - bc \neq 0$ , then  $A$  is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

If  $A$  is an invertible matrix, then we can solve the vector equation  $Ax = b$  by working with  $A^{-1}$ . Indeed, multiplying by  $A^{-1}$  on the left gives

$$\begin{aligned}A^{-1}Ax &= A^{-1}b \\x &= A^{-1}b.\end{aligned}$$

**Theorem.** If  $A$  is an invertible  $n \times n$  matrix, then for all  $b \in \mathbb{R}^n$  the vector equation  $Ax = b$  has a unique solution  $x = A^{-1}b$ .

**Theorem.** If  $A$  and  $B$  are  $n \times n$  invertible matrices, then  $A^T$  and  $AB$  are also invertible and the following hold:

1.  $(A^{-1})^{-1} = A$
2.  $(AB)^{-1} = B^{-1}A^{-1}$
3.  $(A^T)^{-1} = (A^{-1})^T$

**Theorem.** An  $n \times n$  matrix  $A$  is invertible if and only if  $A$  is row equivalent to  $I_n$ , and in this case, any sequence of elementary row operations that reduces  $A$  to  $I_n$  also transforms  $I_n$  into  $A^{-1}$ .

### Algorithm for finding $A^{-1}$

Taking the previous theorem into account, we can row reduce the augmented matrix  $[A \mid I]$ . If  $A$  is row equivalent to  $I$ , then  $[A \mid I]$  is row equivalent to  $[I \mid A^{-1}]$ . Otherwise,  $A$  does not have an inverse.

**Example.** Compute the inverse matrix of

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}.$$

$$\begin{aligned} [A \mid I] &= \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -9/2 & 7 & -3/2 \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & 3/2 & -2 & 1/2 \end{bmatrix} \end{aligned}$$

Thus, the inverse matrix is:

$$A^{-1} = \begin{bmatrix} -9/2 & 7 & -3/2 \\ -2 & 4 & -1 \\ 3/2 & -2 & 1/2 \end{bmatrix}.$$

**Theorem.** (*The Invertible Matrix Theorem*) Let  $A$  be a square  $n \times n$  matrix. Then the following statements are equivalent. That is, for a given  $A$ , the statements are either all true or all false.

- (a)  $A$  is an invertible matrix.
- (b)  $A$  is row equivalent to the  $n \times n$  identity matrix.
- (c)  $A$  has  $n$  pivot positions.
- (d) The equation  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution.
- (e) The columns of  $A$  form a linearly independent set.
- (f) The linear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  is one-to-one (injective).
- (g) The equation  $A\mathbf{x} = \mathbf{b}$  has at least one solution for each  $\mathbf{b} \in \mathbb{R}^n$ .
- (h) The columns of  $A$  span  $\mathbb{R}^n$ .
- (i) The linear transformation  $\mathbf{x} \mapsto A\mathbf{x}$  maps  $\mathbb{R}^n$  onto  $\mathbb{R}^n$  (is surjective).
- (j) There is an  $n \times n$  matrix  $C$  such that  $CA = I$ .
- (k) There is an  $n \times n$  matrix  $D$  such that  $AD = I$ .

(l)  $A^T$  is an invertible matrix.

**Proposition.** Let  $A$  and  $B$  be square matrices. If  $AB = I$ , then  $A$  and  $B$  are both invertible, with  $B = A^{-1}$  and  $A = B^{-1}$ .

**Definition.** A linear transformation  $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is said to be **invertible** if there exists a function  $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$  such that  $S(T(x)) = x$  and  $T(S(x)) = x$  for all  $x$  in  $\mathbb{R}^n$ .

We call  $S$  the **inverse** of  $T$  and write it as  $T^{-1}$ .

**Theorem.** Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be a linear transformation, and let  $A$  be the standard matrix for  $T$ . Then  $T$  is invertible if and only if  $A$  is an invertible matrix. In that case, the linear transformation  $S$  given by  $S(\mathbf{x}) = A^{-1}\mathbf{x}$  is the unique function satisfying the equations from the previous definition.

## Determinants

**Definition.** A determinant is a function that assigns to each square matrix, a real number, that is,  $\det : M_n(\mathbb{R}) \rightarrow \mathbb{R}$ , where  $M_n(\mathbb{R})$  is the set of all square matrices of real numbers. Such function measures how much does a matrix inflate the space when acting as a linear map.

**Example.** If a map given by  $A$  multiplies by 2 the components of a vector, then applying  $A$  over the unit square, will increase its sides by 2, and hence the area of the resulting square will be 4. In this case,  $\det A = 4$ .

Since the determinant is the ratio of expansion of a linear map given by a matrix, we can calculate how much the unit square is expanded by  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . We won't get into the details, but the area of the square produced by  $A$  will be  $ad - bc$ . That's what we call the determinant of a  $2 \times 2$  matrix. This notion can be generalized to  $n \times n$  matrices.

**Theorem.** The **determinant** of an  $n \times n$  matrix  $A$  can be computed by a cofactor expansion across any row or down any column. The expansion across the  $i$ -th row using the cofactors  $C_{ij} = (-1)^{i+j} \det A_{ij}$  is

$$\det A = a_{i1}C_{i1} + \cdots + a_{in}C_{in}.$$

The cofactor expansion down the  $j$ -th column is

$$\det A = a_{1j}C_{1j} + \cdots + a_{nj}C_{nj}.$$

The term  $\det A_{ij}$  is the determinant of the matrix remaining from removing the  $i$ -th row and the  $j$ -th column from  $A$ .

**Theorem.** If  $A$  is a triangular matrix, then  $\det A$  is the product of the entries on the main diagonal of  $A$ .

**Example.** Compute  $\det A$ , where

$$A = \begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$$

The cofactor expansion down the first column of  $A$  has all terms equal to zero except the first. Thus:

$$\det A = 3 \cdot \begin{vmatrix} 2 & -5 & 7 & 3 \\ 0 & 1 & 5 & 0 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & -2 & 0 \end{vmatrix} + 0C_{21} + 0C_{31} + 0C_{41} + 0C_{51}$$

Henceforth, we will omit the zero terms in the cofactor expansion. Next, expand this  $4 \times 4$  determinant down the first column to take advantage of the zeros there. We have

$$\det A = 3 \cdot (2) \cdot \begin{vmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{vmatrix}$$

Following the same procedure, we find that this  $3 \times 3$  determinant is  $-2$ . Hence,

$$\det A = 3 \cdot 2 \cdot (-2) = -12.$$